Title

Project Executable Files – SB Stocks

Project Structure

The SB Stocks application contains two primary modules:

Client

The Client folder contains the React frontend application.

Server

The Server folder contains the Node.js and Express backend application.

Frontend Execution

Open a terminal.

Navigate to the client folder:

```
cd client
```

Install dependencies:

```
npm install
```

Start the frontend:

```
npm run dev
```

The frontend runs on:

```
http://localhost:5173
```

Backend Execution

Open another terminal.

Navigate to the server folder:

```
cd server
```

Install dependencies:

```
npm install
```

Environment Configuration

Create a `.env` file inside the server folder.

Add:

```
MONGO_URI=mongodb://localhost:27017/sbstocks
PORT=8000
JWT_SECRET=your_secure_jwt_secret
STOCK_API_KEY=
ADMIN_EMAIL=admin@example.com
ADMIN_PASSWORD=your_secure_admin_password
```

The actual secret values should not be committed to GitHub.

Database Setup

Ensure MongoDB is running.

If a stock seed script is available, execute:

```
npm run seed
```

Admin Creation

Execute:

```
npm run create-admin
```

Start Backend

Run:

```
npm run dev
```

Alternatively:

```
nodemon index.js
```

The backend runs on:

```
http://localhost:8000
```

Application Execution Order

1. Start MongoDB.
2. Configure backend environment variables.
3. Install backend dependencies.
4. Seed stock data if required.
5. Create the admin account if required.
6. Start the backend server.
7. Install frontend dependencies.
8. Start the frontend development server.
9. Open the application in a web browser.

□ Sample Project Documentation

PROJECT TITLE

SB Stocks – Virtual Stock Market and Paper Trading Platform

ABSTRACT

SB Stocks is a web-based virtual stock trading platform developed to provide users with a risk-free environment for understanding stock market operations.

The application enables users to register, login, browse stocks, analyze historical stock data, create virtual portfolios, and perform simulated buy and sell transactions using virtual funds.

The system maintains transaction and order records and calculates portfolio performance, including current value and profit or loss.

A strategy analysis module enables users to analyze historical market information using basic trading strategies.

The application also contains an administrative module for monitoring users, transactions, orders, and stock information.

SB Stocks is developed using the MERN technology stack. React and Vite are used for frontend development, Node.js and Express.js are used for backend REST APIs, and MongoDB with Mongoose is used for database management.

The application follows the Model-View-Controller architectural pattern to improve modularity, readability, and maintainability.

INTRODUCTION

The stock market is an important component of the financial system. However, beginners may find stock trading difficult because of complex market terminology, financial risk, and limited practical experience.

Learning theoretical concepts alone may not provide sufficient understanding of trading operations.

SB Stocks addresses this problem through paper trading.

Paper trading is the simulation of stock market transactions without using real money.

The application provides users with virtual funds that can be used to simulate stock purchases and sales.

OBJECTIVE

The primary objective of SB Stocks is to create an educational and risk-free stock trading simulation platform.

The project aims to:

- Provide practical stock trading experience.
- Simulate buy and sell operations.
- Enable virtual portfolio management.
- Provide stock data visualization.
- Maintain transaction history.
- Calculate portfolio performance.
- Support historical data analysis.
- Provide administrative monitoring.

EXISTING PROBLEM

Many existing trading applications focus primarily on real financial transactions.

Beginners may hesitate to use these applications because incorrect trading decisions can result in financial loss.

Users also require practical experience to understand portfolio management and transaction processing.

PROPOSED SYSTEM

SB Stocks provides a virtual stock trading system.

Users can create accounts and receive virtual trading funds.

They can browse stocks, view historical information, create portfolios, and simulate trading activities.

The backend validates every transaction before updating the user's balance and portfolio.

SYSTEM MODULES

User Module

Handles registration, login, and user profile information.

Stock Module

Handles stock listings, search, stock details, and historical information.

Portfolio Module

Handles multiple virtual portfolios and holdings.

Transaction Module

Handles virtual buy and sell operations.

Order Module

Maintains order records.

Strategy Analysis Module

Analyzes historical stock data.

Admin Module

Provides administrative monitoring features.

SYSTEM ARCHITECTURE

The application consists of:

- React Frontend.
- Express REST API.
- MongoDB Database.

The frontend communicates with the backend through HTTP requests.

The backend processes requests and accesses MongoDB using Mongoose.

MVC ARCHITECTURE

Model

Defines application data schemas.

Controller

Processes application business logic.

View/Route Layer

Defines REST API endpoints.

The separation improves maintainability and scalability.

DATABASE MODELS

User

Stores user information, role, and virtual balance.

Stock

Stores stock details and historical data.

Portfolio

Stores user portfolios and holdings.

Transaction

Stores buy and sell activities.

Order

Stores stock order records.

AUTHENTICATION

JWT-based authentication is used.

Passwords are securely hashed before database storage.

Protected endpoints validate authentication tokens.

Admin endpoints require role-based authorization.

BUY PROCESS

The system:

1. Validates the user.
2. Validates stock quantity.
3. Retrieves stock price.
4. Calculates purchase value.
5. Checks virtual balance.
6. Validates portfolio ownership.
7. Updates holdings.
8. Deducts virtual funds.
9. Creates a transaction.
10. Creates an order.

SELL PROCESS

The system:

1. Validates the user.
2. Verifies portfolio ownership.
3. Checks stock holdings.
4. Checks available quantity.
5. Calculates sale value.
6. Updates holdings.
7. Updates virtual balance.
8. Records the transaction.
9. Records the order.

ADVANTAGES

- Risk-free learning.
- Practical trading simulation.
- Multiple portfolios.
- Stock analysis.
- Transaction tracking.
- Performance analysis.
- Modular architecture.
- Secure authentication.

LIMITATIONS

- The application does not execute real trades.
- Market information depends on the configured data source.
- Demo market data may be used when an external market API is unavailable.
- Strategy results do not guarantee future performance.

FUTURE ENHANCEMENTS

Future enhancements may include:

- Advanced technical indicators.
- Candlestick charts.
- Stock watchlists.
- Portfolio comparison.
- AI-assisted market education.
- News sentiment analysis.
- More trading strategies.
- Real-time notifications.
- Advanced admin analytics.
- Mobile application support.

CONCLUSION

SB Stocks provides a structured and risk-free environment for learning basic stock trading operations.

The application integrates stock exploration, virtual trading, portfolio management, transaction tracking, historical data analysis, and administrative monitoring.

The MERN technology stack and MVC architecture provide a modular and scalable technical foundation.

The project demonstrates the practical implementation of full-stack web development concepts while addressing the need for an educational paper trading platform.